Espresso

TECHNISCHE UNIVERSITÄT DARMSTADT

Low Level Synthesis: Mini-Tasks Round 1
May 18, 2026

Students:
Sebastian Engel
Amey Wankhede

Tutors:
M.Sc. Katharina Schultheis
Prof. Dr.-Ing Christian Hochberger

1 Basis

A set of benchmarks in BLIF is provided. For comparing the equivalence of two functions, the tool ABC[1] can be used.

2 Task

Implement the logic minimization algorithm ESPRESSO. The program has to be able to read Boolean functions from BLIF files. The algorithm should be realized as it was explained in the lecture, i.e.

- Compute function complement
- Reduce
- Expand (including heuristic for sequence of implicants **and** literals)
- Remove redundancy

The resulting functions should again be written as BLIF files. It is required to use Java/Kotlin as a programming language. Internally, the Positional Cube Notation can be realized using Int Arrays.
To evaluate the implementation, the equivalence of the input and output functions should be verified using ABC. Furthermore, the quality of the implementation should be evaluated by comparing your results with the reference results regarding the number of cubes.

3 Literature

The introduction to ESPRESSO in the lecture is similar to the presentation in *Synthesis and Optimization of Digital Circuits*[2]. Furthermore, *Logic Minimization Algorithms for VLSI Synthesis*[3] is recommended for a more detailed explanation. A documentation of ABC can be found online[4].

4 Formality

A short report (4-10 pages) should be written as explained in the presentation of the mini tasks. A hand in is expected including all files, evaluation and test data, the report as well as other relevant documents. Moreover, a presentation of 10 minutes should be given.

[1] executables of ABC can be found in Moodle, for the source see: https://github.com/berkeley-abc/abc
[2] DE MICHELI, Giovanni, 1994. Synthesis and optimization of digital circuits
[3] BRAYTON, Robert King, 1984. Logic minimization algorithms for VLSI synthesis
[4] https://people.eecs.berkeley.edu/~alanmi/abc/

Sebastian Engel

Amey Wankhede